

PHARMACOKINETICS —

SMART STUDY (By Prof. Dr. Riffat Farooqui)

Instruction: Revise repeatedly (this chapter builds your entire pharmacology base)

1. What is Pharmacokinetics? (Foundation)

Definition: Study of what the body does to a drug

It describes the journey of a drug inside the body

$A \rightarrow D \rightarrow M \rightarrow E$

Absorption

Distribution

Metabolism

Elimination

2. ABSORPTION (Entry of Drug)

Definition:

Movement of drug from site of administration $\rightarrow$ bloodstream

Mechanisms of absorption of drugs from the GIT (Passive diffusion, most common),

Facilitated diffusion, Active transport, exocytosis/endocytosis

Key Factors Affecting Absorption:

Route of administration (oral, IV, etc.)

Blood flow

Surface area (intestine > stomach)

Lipid solubility

pH of drug & body

P-glycoprotein

Bioavailability: Amount of drug reaching circulation

First-pass metabolism: Drug destroyed in liver before reaching blood (Presystemic drug metabolism before reaching circulation)

Example: Oral drugs $\downarrow$ bioavailability due to liver metabolism

3. DISTRIBUTION (Spread of Drug)

Definition:

Movement of drug from blood $\rightarrow$ tissues

Key Factors:

Blood flow to organs (vessel rich organs, high lipophilicity)

Protein binding (albumin)

Lipid solubility

Capillary permeability (slit junctions, lipophilicity, tight junctions)

Definition of V_d and formula

Different compartments in the body where the drug distributed

1) Plasma (low V_d / Heparin)

2) ECF (moderate V_d aminoglycosides)

3) Total body water (large v_d /ethanol)

4. METABOLISM (Drug Breakdown)

Definition:

Chemical conversion of drug → inactive or active form

Main Site:

Liver (most important)

Types:

Phase I → oxidation, reduction, hydrolysis (main enzyme name involved cytochrome p 450)

Phase II → conjugation (makes drug more water-soluble), Name of endogenous substrates of conjugation reactions)

Some drugs become more active after metabolism (prodrugs)

Enzyme induction & inhibition affect drug action (cytochrome p450 inducers and inhibitors name)

Kinetics of metabolism

1st order kinetics definition and example

2nd order kinetics definition and example

5. ELIMINATION (Drug Removal)

Definition:

Removal of drug from body

Main Routes:

Kidney (urine) most important

Bile

Feces

Lungs (gases)

Renal elimination of drugs include Glomerular filtration, proximal tubular secretion via active transport systems for anions and cations, distal tubular reabsorption (weak acidic drugs eliminated by urinary alkalization/ion trapping mechanism)

Important Concepts:

Half-life ($t_{1/2}$): Time for drug concentration to reduce by half

Determines dosing interval

Example: Short half-life → frequent dosing

Definition of steady state concentration of drug

6. ROUTES OF DRUG ADMINISTRATION (go thru page number 5, figure 1.5)

1. Enteral (via GIT)

Oral

Sublingual

Rectal

2. Parenteral (injections)

IV (fastest)

IM
Subcutaneous
3. Topical
Skin
Eye
Inhalation

Step 4: Drug is removed (Elimination)

Must Know Definitions:

Pharmacokinetics
Absorption
Prodrug
Prototype drug
Placebo
Bioavailability
Distribution
Metabolism
Half-life
First-pass metabolism
Protein binding
Liver role in metabolism
Kidney role in excretion

Extract one line questions from every page of the chapter here is the example

Page 1: Overview of Pharmacokinetics

Q: What is pharmacokinetics?
A: Pharmacokinetics is what the body does to a drug.
Q: What is pharmacodynamics?
A: Pharmacodynamics is what the drug does to the body.
Q: What are the four main processes of pharmacokinetics?
A: Absorption, Distribution, Metabolism, and Elimination.
Q: What is absorption?
A: It is the movement of drug from the site of administration into blood.
Q: What is distribution?
A: It is the reversible transfer of drug from blood to tissues.
Q: What is metabolism?
A: It is the chemical transformation of drug mainly in the liver.
Q: What is elimination?
A: It is the removal of drug from the body via urine, bile, or feces.
Q: Why is pharmacokinetics important clinically?
A: It helps in designing proper dose, route, and frequency of drugs.

Page 2: Routes of Drug Administration

Q: What determines the route of drug administration?
A: Drug properties and therapeutic objectives.

Q: What are the major routes of drug administration?

A: Enteral, Parenteral, and Topical.

Q: What is enteral route?

A: Drug administration through the gastrointestinal tract.

Q: Give an example of enteral route.

A: Oral administration.

Q: What is the advantage of oral route?

A: It is convenient and economical.